

Functorial X-CFTs(2D) from Conformal Nets

1.

Axiomatic Approaches to QFT

Analytic
Wightman Axioms
(1964)

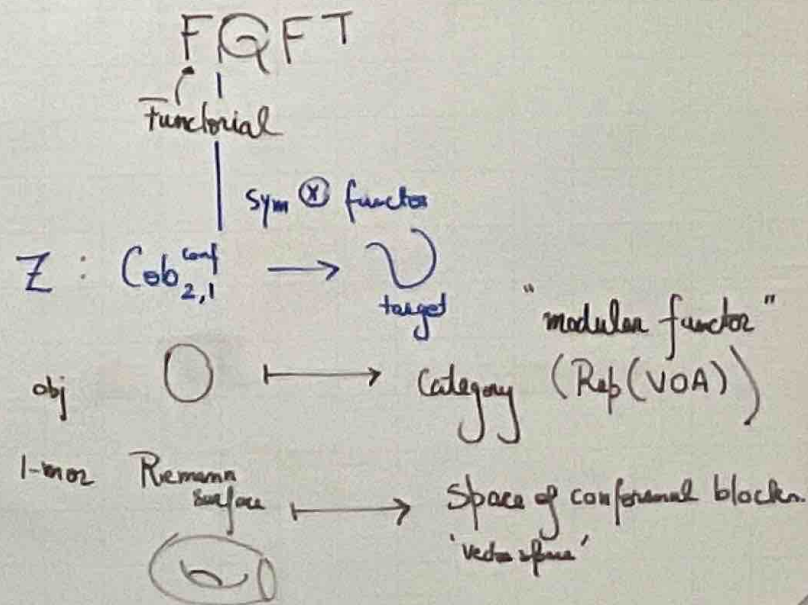
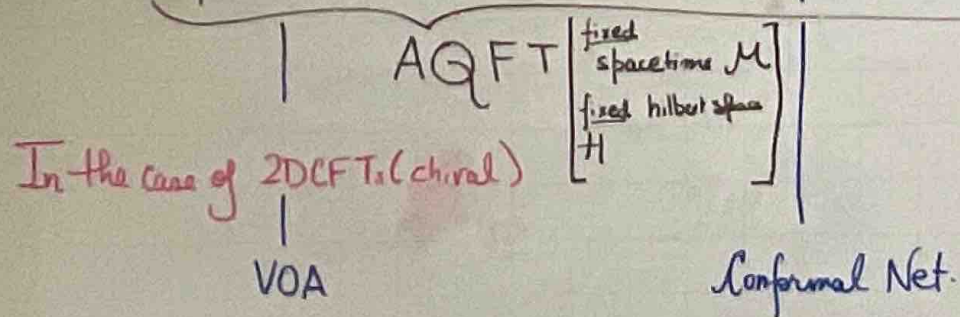
quantum fields
operator valued distributions.

Algebraic
Haag-Kastler Axioms
(1964)

algebra of observables.

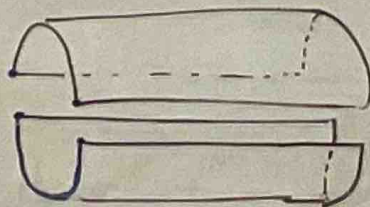
Geometric
Reps of Cobordism Categories
(1988/89)

state spaces and propagators



FQFT

2 obvious generalizations.



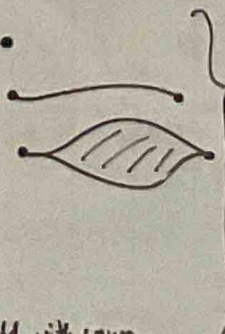
$\text{Cob}_{n,n-1,\dots,0}$

fully extended.

obj	points
1-mor	1-manifolds
2-mor	surfaces
...	...
n-1 mor	n-1 manifold with coun.
n-mor	cobordism "Space-time"
n+1-mor	diffomorphism class of cobordism
n+2-mor	homotopies of diffomorphisms
n+3-mor	homotopies of homotopies
...	...

(∞, n) -category of cobordisms Cob_n .

"slices"



∞

Target

2.

Algebraic "resemble Vect at the top"

Vect
obj: vector space
mor: linear map
 $\otimes$

2-Vect

Lin Cat $\xrightarrow{\text{Alg}}$ Mod (Alg)

obj: linear cat
1-mor: linear fun
2-mor: Nat trans.
algebra
bi-modules
bi-module homomorphisms

3-Vect

Mod (Tens Cat)

obj: $\otimes$ categories
1-mor: bi-module category
2-mor: bi-linear functors
3-mor: natural trans.

with good adjoints there are
equivalent

Unitary

Hilb

2-Hilb

W^* -Cat

Mod (VNAlg)

3-Hilb "Work in Progress"

Mod (Bicommutat Cat)

W^* -Categories (talked about in 1985 Ghaz, Lima, Roberts) 3.

Def: A $*$ -category $\mathcal{C}$ is a $\mathbb{C}$ -linear category with a dagger structure $*$: $\text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(Y, X)$ which is $\mathbb{C}$ -antilinear and $f^{**} = f$, $(g \circ f)^* = f^* \circ g^*$.

Def: $\mathcal{C}^{\oplus}$ to be direct sum completion of $\mathcal{C}$.
 obj : formal direct sum mor : $\text{Hom}_{\mathcal{C}^{\oplus}}(\bigoplus_i X_i, \bigoplus_j X_j) = \bigoplus_{ij} \text{Hom}_{\mathcal{C}}(X_i, X_j)$

Def: A $*$ -category is a C^* -category if $\text{End}_{\mathcal{C}}(X)$ is a C^* -algebra $\forall X \in \text{obj}(\mathcal{C}^{\oplus})$

W^* -Category if $\text{End}_{\mathcal{C}}(X)$ is a vN algebra $\forall X \in \text{obj}(\mathcal{C}^{\oplus})$

A W^* -functor $F : \mathcal{C}_1 \rightarrow \mathcal{C}_2$ induces normal homomorphism $\text{End}_{\mathcal{C}_1}(X) \rightarrow \text{End}_{\mathcal{C}_2}(F(X)) \forall X \in \mathcal{C}_1$.

$\overline{\mathcal{C}}$: conjugate W^* -category obj : objs of $\mathcal{C}$
 mor : complex conjugate morphism spaces of $\mathcal{C}$.

W^* -Categories are 2-hilbert spaces.

def : $F : \mathcal{C}_1 \rightarrow \mathcal{C}_2$ define $F^* : \mathcal{C}_2 \rightarrow \mathcal{C}_1$ such that $\langle m_1, F^*(m_2) \rangle_{\mathcal{C}_1} = \langle F(m_1), m_2 \rangle_{\mathcal{C}_2} \forall m_1 \in \mathcal{C}_1, m_2 \in \mathcal{C}_2$

Properties:

(a) Every W^* -category admits a canonical Hilb-valued inner product

$$\langle -, - \rangle_c : \bar{\mathcal{C}} \times \mathcal{C} \rightarrow \text{Hilb}$$

$$\langle X, Y \rangle_c := p_Y L^2(\text{End}_c(X \oplus Y)) p_X \quad \text{where } p_X, p_Y \in \text{End}_c(X \oplus Y)$$

in particular $\langle X, X \rangle_c = \underbrace{L^2 \text{End}_c(X)}_{\text{Hilbert space}}$

, comes equipped with an isomorphism

$$\Psi_{X,Y} : \langle X, Y \rangle_c \xrightarrow{\sim} \overline{\langle Y, X \rangle_c}$$

Adjective

(a) Completely additive : Idempotent complete, admits all orthogonal direct sums.

(b) admits a generator : $\exists$ an object $A \neq 0$ st. $\text{Hom}_{\mathcal{C}}(A, X) \neq 0$ for all $X \neq 0$.

Theorem: Let $\mathcal{C}$ be a completely additive W^* -category that admits a generator then

(i) $\mathcal{C} \cong R\text{-Mod}$ for some unital R

(ii) $\mathcal{C} \cong \text{Func}_{W^*}(\bar{\mathcal{C}}, \text{Hilb})$

where this isomorphism comes from

$$A \longmapsto \langle -, A \rangle_c$$

(2-Riesz Representation theorem).

Absorbing Objects.

Let $\mathcal{T}$ is a tensor W^* -category.

Def. A weakly-absorbing object is a non-zero divisor $\Omega \in \mathcal{T}$ such that $\forall$ non-zero divisors $X \in \mathcal{T}$

$$\begin{array}{l} \text{(left)} \quad \Omega \otimes X \simeq \bigoplus_I \Omega \\ \text{(right)} \quad X \otimes \Omega \simeq \bigoplus_I \Omega \end{array} \quad \text{for some indexing set } I.$$

Denote $\mathcal{T}^{ab}$ as the idempotent completion of full-subcategory of weakly-absorbing objects ($\mathcal{T}^{ab} \subseteq \mathcal{T}$) is the absorbing ideal.

Th. Let $\mathcal{M}, \mathcal{N}$ be W^* -module categories over $\mathcal{T}$.

Let $\Omega \triangleright_{\mathcal{M}} _ : \mathcal{M} \rightarrow \mathcal{M}$, $\Omega \triangleright_{\mathcal{N}} _ : \mathcal{N} \rightarrow \mathcal{N}$ be faithful and dominant.

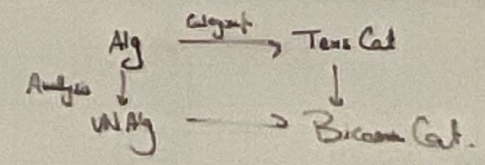
Then $\text{Hom}_{\mathcal{T}\text{-Mod}}(\mathcal{M}, \mathcal{N}) \simeq \text{Hom}_{\mathcal{T}^{ab}}(\mathcal{M}, \mathcal{N})$.

Cor:

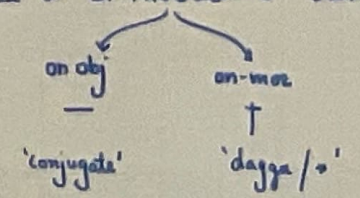
half-braiding of X with $\mathcal{T}$ is equivalent to half-braiding of X with $\mathcal{T}^{ab}$

Bicommutant Categories

Motivation: categorified vN-algebras.



Def: A bicommutant category $\mathcal{T}$ is a bi-involutive tensor category whose underlying dagger category is a



completely additive W^* -category with a generator that admits weakly absorbing objects.

$$L: \mathcal{T} \rightarrow \text{End}_{W^*cat}(\mathcal{T}^{abs}) \quad R: \mathcal{T} \rightarrow \text{End}_{W^*cat}(\mathcal{T}^{abs})$$

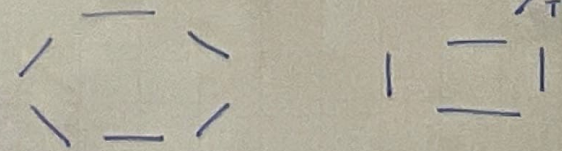
are bi-involutive $\otimes$ functors

i.e they come with isomorphisms.

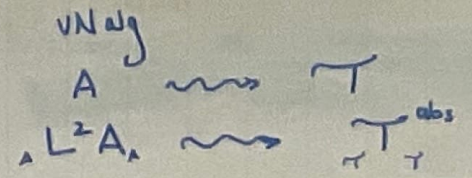
$$\begin{aligned} \langle L(X)(\Omega_1), \Omega_2 \rangle_{\mathcal{T}^{abs}} &\cong \langle \Omega_1, L(\bar{X})(\Omega_2) \rangle_{\mathcal{T}^{abs}} \\ \langle R(X)(\Omega_1), \Omega_2 \rangle_{\mathcal{T}^{abs}} &\cong \langle \Omega_1, R(\bar{X})(\Omega_2) \rangle_{\mathcal{T}^{abs}} \end{aligned}$$

$$\begin{aligned} \langle X \otimes \Omega_1, \Omega_2 \rangle &\cong \langle \Omega_1, \bar{X} \otimes \Omega_2 \rangle \\ \langle \Omega_1 \otimes X, \Omega_2 \rangle &\cong \langle \Omega_1, \Omega_2 \otimes \bar{X} \rangle \end{aligned}$$

which satisfy



And, the left and Right actions of $\mathcal{T}$ on $\mathcal{T}^{abs}$ are commutants of each other $[Im(L)]' \cong Im R$.



(on a circle S^1)

Conformal Nets

7.

data : H_0

$U : \text{Diff}^+(S^1) \rightarrow \mathcal{PB}(H_0)$ projective unitary representation

$\forall I \subsetneq S^1, \mathcal{A}(I) \subseteq \mathcal{B}(H_0)$ a VN-algebra.

Axioms

- $I \subseteq J \implies \mathcal{A}(I) \hookrightarrow \mathcal{A}(J)$
- $I \cap J = \emptyset \implies [\mathcal{A}(I), \mathcal{A}(J)]_{\mathcal{B}(H_0)} = 0$

$\forall g \in \text{Diff}^+(S^1), \forall I \subsetneq S^1$, let $V \in \mathcal{B}(H_0)$ represent $U(g)$
 $V \mathcal{A}(I) V^* = \mathcal{A}(g \cdot I)$.

$\exists \Omega \in H_0$ (unique up to scaling) a vacuum vector which is proj. preserved by U and cyclic for $\bigcup_{I \subsetneq S^1} \mathcal{A}(I)$.

Can be made "Co-ordinate free".

$\mathcal{A} : \text{INT} \longrightarrow \text{VNAlg}$
 $I \longmapsto \mathcal{A}(I)$

isoturbation preserving dif $\longrightarrow$ homomorphisms of VNAlg.

Input $\xrightarrow{I}$, Conformal Net A
 $A(I)$ - vN algebra.

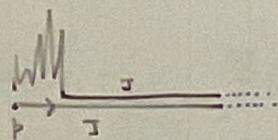
$$T_A^+(I)$$

'left-localized endomorphism'

$$\text{obj: } \rho: A(I) \rightarrow A(I)$$

$$\text{s.t. } \exists J, \partial I \subset J$$

$$\rho|_{A(J)} = \text{id}_{A(J)}$$



$$\text{mor: } a_{\rho, \sigma}: \rho \rightarrow \sigma$$

$$a_{\rho, \sigma} \in A(K), \text{ s.t. } \partial I \not\subset K \text{ (denote } A(I)^{\leftarrow})$$

$$\text{s.t. } a \rho(x) = \sigma(x) a \quad \forall x \in A(I).$$

$$\text{dagger: } (a_{\rho, \sigma})^\dagger = a_{\sigma, \rho}^*$$

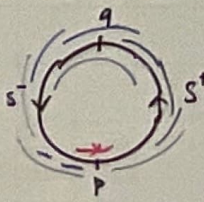
$$\textcircled{X} \quad \begin{array}{l} \text{on obj.} \quad \rho \otimes \sigma = \sigma \circ \rho \\ \text{on map} \quad a_{\rho, \sigma} \otimes a_{\rho', \sigma'} = (\sigma'(a) a')_{\rho \otimes \rho', \sigma \otimes \sigma'} \end{array}$$

Similarly $T_A^-(I) \xrightarrow{I}$ right-localized endomorphism

$$\text{Th: } T_A^{+-}(i): T_A^{+-}(I) \xrightarrow{\sim} T_A^{+-}(I')$$

$\textcircled{X}$ equivalence of categories.

Solitons

Input S^1 with s^-  , conformal Net A

$$T_p(S)$$

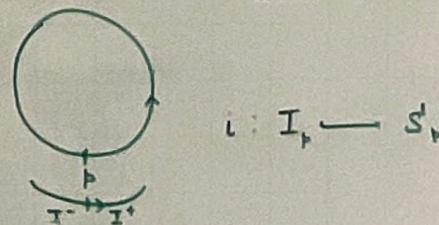
$$\text{obj: Hilbert space with action } (H, \{\alpha_J: A(J) \rightarrow \mathcal{B}(H)\}_{J \subset S^1, p \notin J})$$

mor: bounded linear maps that intertwine actions.

involution of objects: $\overline{H}$ (complex conjugate Hilbert space)

$$\textcircled{X} \text{ is given as } H_1 \otimes H_2 := \text{Hom}_{A(S^1)\text{-Mod}}(H_1, H_2) \otimes_{A(S^1)} H_2.$$

Thm 1:



gives an equivalence of monoidal categories

$$T_A^+(I^+) \simeq T_p(S) \simeq T_A^-(I^-)^{\text{op}}$$

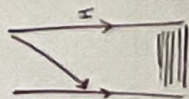
In spirit of

$$\text{Rep}(A) \simeq \text{DHR} \xrightarrow{\sim} \text{End}(A) \simeq \text{Bim}(A)$$

Thm 2: $\mathcal{T}_A^+(I)$ is a bicommutant category.

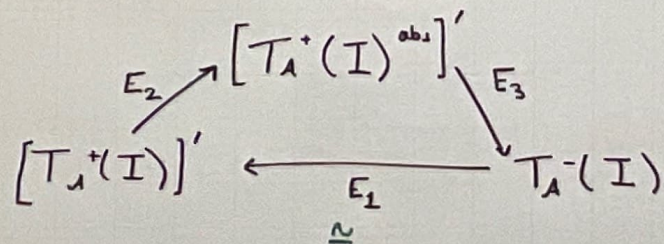
9.

Step 1: Identify the absorbing object.



Lemma: $\Phi: I \rightarrow I$ is a non-surjective embedding such that $\exists J \subset I, \partial+I \in J$ and $\Phi(J) = \text{id}_J$
 then $\mathcal{A}(\Phi)$ is an absorbing object in $\mathcal{T}_A^+(I)$

Step 2: Show $\mathcal{T}_A^-(I) \simeq [\mathcal{T}_A^+(I)]'_{\text{End}(\mathcal{A}(I))}$



• $E_3 \circ E_2 \circ E_1 \simeq \text{id}_{\mathcal{T}_A^-(I)}$ *

• E_2 is fully-faithful.
 and $E_2 \circ E_1 \circ E_3 \simeq \text{id}_{[\mathcal{T}_A^+(I)^{abs}]'}$ *

} $E_1 \circ E_3 \circ E_2 \simeq \text{id}_{[\mathcal{T}_A^+(I)]'}$ *

Thm 3: $\Sigma^{\text{Drinfeld}}(\mathcal{T}_A^+(I)) = \text{Rep}(\mathcal{A})$

Examples (WZW case) $\Sigma^{\text{Drinfeld}}(\text{Rep}^*(\Omega G)) \simeq \text{Rep}^*(LG)$

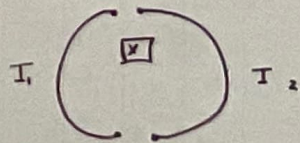
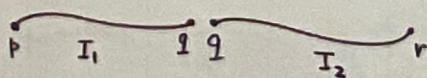
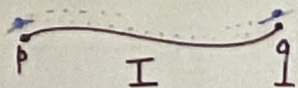
(Virasoro case) $\Sigma^{\text{Drinfeld}}(\text{Rep}_c(\text{DHF}(S', p))) \simeq \text{Rep}_c(\text{DHF}(S'))$

$\text{Cob}_{2,1,0}^{\text{conf}}$

obj

 I^+

1-mor



Bicom Cat

$$\mathcal{T}_p := \mathcal{T}_a^+(I^+)$$

$$\mathcal{T}_p \mathcal{A}(I)\text{-Mod}_{\mathcal{T}_q}$$

 W^* -bi-modules over bi-commutant categories

$$\mathcal{A}(I)\text{-Mod}_{\mathcal{T}_q \oplus \mathcal{T}_p^+}$$

$$\mathcal{T}_p \mathcal{A}(I_1)\text{-Mod}_{\boxed{\times}_{\mathcal{T}_q}} \mathcal{A}(I_2)\text{-Mod}_{\mathcal{T}_r} \simeq \mathcal{T}_p \mathcal{A}(I_1 \cup I_2)\text{-Mod}_{\mathcal{T}_r}$$

$$\left[\text{Rep}(\mathcal{A}(I_1 \cup I_2)) \right]^{\text{abs.}}$$

$$\left[\text{in rigid rational case } \text{Rep} A^{\text{abs}} \simeq \text{Rep} A \right]$$

↓
"categorified Connes fusion"

$$\mathcal{A}(I_1)\text{-Mod}_{\boxed{\times}_{\mathcal{T}_2}} \text{Hom}_{\mathcal{T}_1\text{-Mod}}(\mathcal{T}_2^{\text{abs}}, \mathcal{A}(I_2)\text{-Mod})$$

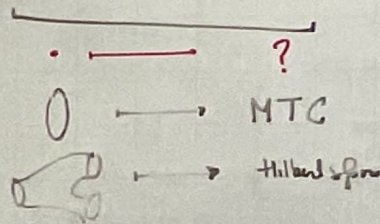
"Work in progress."

Final Thoughts.

11.

1. Aim to construct fully-extended functorial χ -CFT [topological at levels 0, 1].

2. Relationship between 2D CFTs and 3D TQFTs in non-rational cases. [Virasoro TQFT]



1. $Z(\bigcirc)$ is the center of $Z(\cdot)$
2. $Z(\cdot)$ is a fully-dualizable obj in the target.
3. fd obj in TensorCat are fusion categories
4. $\exists$ MTCs that do not arise as Drinfeld (finite semi-simple) etc. centres of any fusion categories

So, we make the target 'slightly' bigger st

$\exists$ object in BicomCat whose centres are assigned to S^1 in general.

3. Physical Interpretation of T_p in terms of defects

(we know that it contains more information than just topological defects)

4. Develop the theory for Bicommutable Categories and modules.

as rigorously as for tensor categories.